

JEE MAIN REVISION MASTER: ELECTROMAGNETIC INDUCTION

Subject: Physics | Target Scope: JEE Main Previous Years | Empirical Base: 153 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Magnetic Flux & Faraday-Lenz Law	92 PYQs	Core Principle
Very High	Self Inductance & RL Transient Circuits	26 PYQs	Inductance & Circuits
Very High	Motional EMF (Translational & Rotational)	25 PYQs	Motional Induction
Very High	Mutual Inductance & Coupled Coils	17 PYQs	Coupling
Standard	Induced Electric Field & Eddy Currents	1 PYQs	Field Theory

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Magnetic Flux & Faraday-Lenz Law	$\Phi_B = \vec{B} \cdot \vec{A} = B A \cos\theta$, $\mathcal{E} = -\frac{d\Phi_B}{dt} = -N \frac{d\Phi_B}{dt}$, $q = \frac{\Delta \Phi}{R}$
Self Inductance & RL Transient Circuits	$\mathcal{E} = -L \frac{di}{dt}$, $L = \mu_0 n^2 A l$, $U = \frac{1}{2} L I^2$, $\tau = \frac{L}{R}$, $I(t) = \frac{V}{R} (1 - e^{-t/\tau})$
Motional EMF (Translational & Rotational)	$\mathcal{E} = B v L$, $I = \frac{B v L}{R}$, $F_{\text{mag}} = \frac{B^2 L^2 v}{R}$, $\mathcal{E}_{\text{rot}} = \frac{1}{2} B \omega L^2$
Mutual Inductance & Coupled Coils	$\mathcal{E}_2 = -M \frac{di_1}{dt}$, $M = \mu_0 n_1 n_2 (\pi r_{\text{inner}}^2) l$, $M = k \sqrt{L_1 L_2}$ ($0 \leq k \leq 1$)
Induced Electric Field & Eddy Currents	$\oint \vec{E} \cdot d\vec{l} = -\frac{d\Phi_B}{dt}$, $E(r) = \frac{r}{2} \frac{dB}{dt}$ ($r \leq R$), $E(r) = \frac{R^2}{2r} \frac{dB}{dt}$ ($r > R$)

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Magnetic Flux & Faraday-Lenz Law (Very High Yield — 92 PYQs)

Core Principle: Magnetic flux $\Phi_B = \vec{B} \cdot \vec{A} = B A \cos\theta$. Induced EMF $\mathcal{E} = -(\mathrm{d}\Phi_B / \mathrm{d}t) = -N (\mathrm{d}\Phi_B / \mathrm{d}t)$. Lenz's law dictates that induced currents oppose the change in magnetic flux.

■ **Caution / JEE Pitfall:** 1. Total induced charge flown $\Delta q = (\Delta\Phi_B / R)$ is independent of the rate or time interval of flux change! 2. Angle θ is measured from the surface normal $\vec{A}$, not the coil plane.

■ **Speed Trick:** When an N -turn coil rotates with angular velocity ω in field B : $\mathcal{E}(t) = N B A \omega \sin(\omega t)$, with peak EMF $\mathcal{E}_0 = N B A \omega$.

Representative Source PYQs:

- **[2026 Evening Shift] Q#1 (MCQ):** "A 30 cm long solenoid has 10 turns per cm and area of 5cm^2 . The current through the solenoid coil varies from 2 A to 4 A in 3.14 s. $T = \dots$ " **[Ans: B]**
- **[2026 Evening Shift] Q#1 (MCQ):** "Suppose a long solenoid of 100 cm length, radius 2 cm having 500 turns per unit length, carries a current $I = 10\sin(\omega t)$ A, where $\omega = \dots$ " **[Ans: N/A]**
- **[2026 Morning Shift] Q#1 (MCQ):** "Three identical coils C_1, C_2 and C_3 are closely placed such that they share a common axis. C_2 is exactly midway. C_1 ca..." **[Ans: B]**

3.2 Self Inductance & RL Transient Circuits (Very High Yield — 26 PYQs)

Core Principle: Self-inductance $L = (N\Phi / I)$, induced back-EMF $\mathcal{E} = -L(\mathrm{d}I / \mathrm{d}t)$. Solenoid $L = \mu_0 n^2 A l$. RL circuit current growth: $I(t) = (V / R)(1 - e^{-t/\tau})$ with inductive time constant $\tau = (L / R)$.

■ **Caution / JEE Pitfall:** At $t = 0^+$ (immediately after switch closing), an unenergized inductor acts as an open circuit ($R \rightarrow \infty$). At $t \rightarrow \infty$ (steady state), an ideal inductor acts as a zero-resistance short wire.

■ **Speed Trick:** Magnetic energy stored: $U_B = (1/2) L I^2$; energy density in space: $u_B = (B^2 / 2\mu_0)$. Inductors resist sudden changes in current, not voltage.

Representative Source PYQs:

- **[2026 Morning Shift] Q#1 (Numerical):** "Inductance of a coil with 10^4 turns is 10 mH and it is connected to a dc source of 10 V with internal resistance of 10Ω . The en..." **[Ans: 20]**
- **[2026 Morning Shift] Q#3 (MCQ):** "Match the LIST-I with LIST-II List-I List-II A. Magnetic induction I . $\dots$ " **[Ans: D]**
- **[2025 Morning Shift] Q#1 (MCQ):** "Consider I_1 and I_2 are the currents flowing simultaneously in two nearby coils 1 & 2, respectively. If $L_1 =$ self inductance of coil 1, $M = \dots$ " **[Ans: N/A]**

3.3 Motional EMF (Translational & Rotational) (Very High Yield — 25 PYQs)

Core Principle: Motional EMF in a conductor moving through field $\vec{B}$: $\mathcal{E} = B v L$. For a rod rotating about one pivot with angular speed ω : $\mathcal{E}_{\text{rot}} = (1/2) B \omega L^2$.

■ **Caution / JEE Pitfall:** 1. Mechanical power delivered by external pulling force $P_{\text{(mech)}} = F_{\text{(ext)}} v$ exactly matches electrical Joule dissipation $I^2 R = (B^2 L^2 v^2 / R)$. 2. For an arbitrary curved wire, replace it with the straight vector connecting its endpoints.

■ **Speed Trick:** Direction of induced EMF follows $\vec{v} \times \vec{B}$. Positive charges accumulate at the end pointed to by $\vec{v} \times \vec{B}$.

Representative Source PYQs:

- [2026 Morning Shift] Q#2 (MCQ): "A 20 m long uniform copper wire held horizontally is allowed to fall under the gravity ($g=10\text{m/s}^2$) through a uniform horizontal magnetic..." [Ans: N/A]
- [2026 Morning Shift] Q#3 (MCQ): "A 1 m long metal rod AB completes the circuit as shown in figure. The area of circuit is perpendicular to the magnetic field of 0.10 T. If ..." [Ans: B]
- [2026 Morning Shift] Q#3 (Numerical): "A simple pendulum made of mass 10 g and a metallic wire of length 10 cm is suspended vertically in a uniform magnetic field of 2 T. The mag..." [Ans: 100]

3.4 Mutual Inductance & Coupled Coils (Very High Yield — 17 PYQs)

Core Principle: Coupled flux in secondary coil $\Phi_2 = M I_1$, induced EMF $\mathcal{E}_2 = -M(dI_1 / dt)$. Coaxial solenoids $M = \mu_0 n_1 n_2 (\pi r_{\text{(inner)}}^2) l$. Coupling coefficient $M = k\sqrt{L_1 L_2}$ ($0 \leq k \leq 1$).

■ **Caution / JEE Pitfall:** For concentric coaxial coils, always use the cross-sectional area of the INNER coil ($r_{\text{(inner)}}$), because the magnetic field of the outer coil only penetrates the inner core.

■ **Speed Trick:** Reciprocity theorem holds: $M_{12} = M_{21}$ regardless of differing geometry or number of turns.

Representative Source PYQs:

- [2026 Evening Shift] Q#1 (MCQ): "A circular current loop of radius R is placed inside square loop of side length L ($L \geq R$) such that they are co-planar and their cente..." [Ans: D]
- [2026 Morning Shift] Q#2 (Numerical): "In the given circuit below inductance values of L_1, L_2 and L_3 are same. The magnetic energy stored in the entire circuit is $\$(U_{\dots})$ " [Ans: 6]
- [2024 Morning Shift] Q#1 (Numerical): "A small square loop of wire of side l is placed inside a large square loop of wire of side L ($L=l^2$). The loops are coplanar and their..." [Ans: 128]

3.5 Induced Electric Field & Eddy Currents (Standard Yield — 1 PYQs)

Core Principle: A time-varying magnetic field generates a non-conservative, circular electric field: $\oint \vec{E} \cdot d\vec{l} = -(d\Phi_B / dt)$. Inside cylinder ($r \leq R$): $E(r) = (r/2)(dB / dt)$; outside ($r > R$): $E(r) = (R^2 / 2r)(dB / dt)$.

■ **Caution / JEE Pitfall:** Induced electric fields are non-electrostatic: field lines form continuous closed loops and work done along a closed circuit is non-zero ($\oint \vec{E} \cdot d\vec{l} = \mathcal{E} \neq 0$)!

■ **Speed Trick:** Eddy currents and hysteresis losses in transformer cores are minimized by using thin laminated sheets of silicon steel insulated by varnish.

Representative Source PYQs:

- [2022 Evening Shift] Q#2 (MCQ): "The electric current in a circular coil of 2 turns produces a magnetic induction B at its centre. The coil is unwound and is rewound into ..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 8th April Evening Shift (Q#1)

A 30 cm long solenoid has 10 turns per cm and area of 5cm^2 . The current through the solenoid coil varies from 2 A to 4 A in 3.14 s. The e.m.f. induced in the coil is $\alpha \times 10^{-5}\text{V}$. The value α is ____.

- (A) 60
- (B) 12 [Correct]
- (C) 120
- (D) 34

Correct Answer: B

Solution Analysis: According to Faraday's Law, a change in the magnetic flux (Φ) through a coil induces an electromotive force (e.m.f.) in that coil. The magnitude of the induced e.m.f. (ϵ) is given by: $\epsilon = N(d\Phi / dt)$ Where N is the total number of turns. For a solenoid, the magnetic flux Φ through each turn is $\Phi = B \cdot A$, where B is the magnetic field inside the solenoid and A is the cross-sectional area. The magnetic field B inside an ideal solenoid is: $B = \mu_0 n I$ Where μ_0 is the permeability of free space ($4\pi \times 10^{-7}\text{T} \cdot (\text{m} / \text{A})$), n

Benchmark Problem 2: JEE Main 2026 (Online) 2nd April Evening Shift (Q#1)

A circular current loop of radius R is placed inside square loop of side length L ($L \geq R$) such that they are co-planar and their centers coincide. The permeability of free space is μ_0 . The mutual inductance between circular loop and square loop is ____.

- (A) $2\sqrt{2} \frac{\mu_0}{4\pi} L^2 R$
- (B) $\sqrt{2} \frac{\mu_0}{4\pi} L^2 R$
- (C) $\sqrt{2} \frac{\mu_0}{4\pi} R^2 L$
- (D) $2\sqrt{2} \frac{\mu_0}{4\pi} R^2 L$ [Correct]

Correct Answer: D

Solution Analysis: Mutual inductance M is defined by the flux Φ linked with one loop due to a current I in the other: $\Phi = MI$. It is easier to calculate the magnetic field produced by the larger square loop at the centre where the small circular loop is located. A square loop consists of four wires of length L . The distance from the centre to each wire is $r = (L / 2)$. The magnetic field at the centre due to one wire of length L is: $B_{\text{wire}} = \frac{\mu_0}{4\pi} \frac{I}{r} (\sin\theta_1 + \sin\theta_2)$ For a square centre, $\theta_1 = \theta_2 = 45^\circ$ and $r = (L / 2)$. $B_{\text{wire}} = \frac{\mu_0 I}{4\pi r} (1 + 1) = \frac{\mu_0 I}{2\pi r}$

Benchmark Problem 3: JEE Main 2026 (Online) 22nd January Morning Shift (Q#1)

Inductance of a coil with 10^4 turns is 10 mH and it is connected to a dc source of 10 V with internal resistance of 10Ω . The energy density in the inductor when the current reaches $((1 / e))$ of its maximum value is $\alpha \pi \times 10^{-2}\text{J/m}^3$. The value of α is ____ . ($\mu_0 = 4\pi \times 10^{-7}\text{Tm/A}$).

Correct Answer: 20

Solution Analysis: When an inductor is connected to a DC voltage source (V) with a resistance (R), it initially opposes the flow of current. Over time, the current increases until it reaches a steady, maximum state. Once the current is steady, the inductor acts like an ideal wire (zero resistance). Therefore, the maximum current (i_{max}) is determined by Ohm's Law: $i_{\text{max}} = (V / R)$ For given circuit: Voltage $V = 10\text{V}$ Resistance $R = 10\Omega$ $i_{\text{max}} = (10 / 10) = 1\text{A}$ The current in the circuit is $(1 / e)$ time the maximum value, $i = i_{\text{max}} \times ((1 / e)) = 1 \times (1 / e)$

Benchmark Problem 4: JEE Main 2026 (Online) 21st January Morning Shift (Q#3)

A 1 m long metal rod AB completes the circuit as shown in figure. The area of circuit is perpendicular to the magnetic field of 0.10 T . If the resistance of the total circuit is 2Ω then the force needed to move the rod towards right with constant speed (v) of 1.5m/s is ____ N.

- (A) 5.7×10^{-2}
- (B) 7.5×10^{-3} [Correct]
- (C) 5.7×10^{-3}
- (D) 7.5×10^{-2}

Correct Answer: B

Solution Analysis: When a conductor moves through a uniform magnetic field, it experiences motional EMF. For a rod of length l moving with velocity v perpendicular to a magnetic field B , the induced emf is : $\varepsilon = Bvl$ The given values are, $B=0.10\text{T}$ $v=1.5\text{m/s}$ $l=1\text{m}$ Putting the values, $\varepsilon=(0.10) \times (1.5) \times (1)=0.15\text{Volts}$ Using Ohm's Law, where R is the total resistance of the circuit: $I=(\varepsilon / R)$ Given $R=2\Omega$: $I=(0.15 / 2)=0.075\text{Amperes}$ A current-carrying wire in a magnetic field experiences a force given by: $F_{(m)}=IlB\sin(\theta)$ Since the rod is per